

Leo Shaw

Artificial intelligence and mechanical engineering for real-world autonomy, manufacturing systems, and grounded technical reasoning.

Downloadable portfolio companion to the live site. All section links below remain clickable in the PDF.

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Overview

I build systems that sense, decide, and act in physical environments. My work spans semiconductor equipment, robotic sorting cells, reinforcement learning planners, document intelligence pipelines, and grounded RAG systems for manufacturing knowledge.

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Selected Research And Projects

NEXA: Small Language Models for Manufacturing Process Design

Research | 2026 - Present

A grounded manufacturing AI stack combining Nemotron OCR ingestion, handwritten log reconstruction, Q/A/context generation, diagnostic RAG, and OpenWebUI integration.

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Autonomous Driving Local Planner with Reinforcement Learning

January 2026 - May 2026

A CARLA planning framework for merges, left turns, and cut-ins that compares classical planning against PPO-based local trajectory generation under a fixed PID control stack.

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Real-Time Conveyor-Belt Object Detection and Robotic Sorting

January 2026 - May 2026

A final integrated sorting system using computer vision, homography calibration, and a vacuum-equipped delta robot to remove aluminum cans from a moving conveyor.

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More Project Work

Autonomous Drone Pathing with Reinforcement Learning

Spring 2026

A MuJoCo quadrotor navigation project comparing PPO, SAC, and model-based RL, including goal-hover shaping and wind-robustness experiments.

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Seeing Through Occlusion

January 2026 - May 2026

A video instance segmentation project on OVIS that models object motion through occlusion with multimodal LSTMs, temporal attention, and an occlusion-aware memory bank.

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Experience Highlights

- **Ulvac Technologies - Field Service Engineer I** | Santa Clara, CA | Oct 2024 - Aug 2025
Rebuilt and upgraded semiconductor tools across six high-volume fabs, executed 30+ retrofit projects across vacuum, RF, PLC, HMI, and motion subsystems, and improved uptime, process stability, and fault-diagnosis speed.
- **Carnegie Mellon University - Graduate Research Assistant** | Pittsburgh, PA | 2026 - Present
Developing manufacturing-focused small language model workflows, document pipelines, and grounded reasoning systems that convert process documentation into retrieval-ready technical knowledge.
- **Paltorc - Mechanical Engineering Intern** | Milpitas, CA | May 2022 - Aug 2022
Integrated an e-bike control module with rider-facing software and improved telemetry, diagnostics, and overall system reliability.

Quick Links

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[NEXA research page](#)

[GitHub profile](#)

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